VIVEKANANDHA SAINIK COACHING CENTER

AISSEE Class 9 Mock Test - Mock 3 Solution Key & Explanations

Total Questions: 150 | **Full Marks:** 400 | **Subject:** All Sections (Maths, English, Science, SS, Intelligence)

SECTION A: MATHEMATICS (Q1 - Q50)

Q1. Find the additive inverse of -7/12.

A) 7/12 **B)** -7/12 **C)** 12/7 **D)** -12/7

Correct Answer: Option (A) — 7/12

Step-by-Step Solution: The additive inverse of a number x is -x. Hence, -(-7/12) = 7/12.

Q2. Solve for x: 5x - 7 = 2x + 8.

A) 3 **B)** 5 **C)** 4 **D)** 2

Correct Answer: Option (B) — 5

Step-by-Step Solution: Subtract 2x from both sides: 3x - 7 = 8. Add 7: 3x = 15 => x = 5.

Q3. The sum of interior angles of a convex polygon with 8 sides is:

A) 1080° **B)** 900° **C)** 1260° **D)** 1440°

Correct Answer: Option (A) — 1080°

Step-by-Step Solution: Sum of interior angles = (n - 2) * 180° = (8 - 2) * 180° = 6 * 180° = 1080°.

Q4. A die is thrown once. What is the probability of getting a prime number?

A) 1/6 **B)** 1/2 **C)** 1/3 **D)** 2/3

Correct Answer: Option (B) — 1/2

Step-by-Step Solution: Prime numbers on a die are {2, 3, 5}. Total outcomes = 6. Probability = 3/6 = 1/2.

Q5. What is the smallest square number divisible by 6, 9 and 15?

A) 100 **B)** 400 **C)** 900 **D)** 225

Correct Answer: Option (C) — 900

Step-by-Step Solution: LCM of 6, 9, 15 is 90 = 2 * 3^2 * 5. To make it a perfect square, multiply by 2 * 5 = 10. Number = 900.

Q6. Find the cube root of 13824.

A) 22 **B)** 24 **C)** 26 **D)** 28

Correct Answer: Option (B) — 24

Step-by-Step Solution: 13824 = 2^9 * 3^3 = (2^3 * 3)^3 = (8 * 3)^3 = 24^3. Thus, cube root is 24.

Q7. An article purchased for Rs 250 is sold for Rs 300. Find the profit percentage.

A) 15% **B)** 20% **C)** 25% **D)** 10%

Correct Answer: Option (B) — 20%

Step-by-Step Solution: Profit = SP - CP = 300 - 250 = Rs 50. Profit % = (50 / 250) * 100 = 20%.

Q8. Multiply (2x + 3y) and (3x - 2y).

A) $6x^2 + 5xy - 6y^2$ **B)** $6x^2 - 5xy - 6y^2$ **C)** $6x^2 + 6y^2$ **D)** $6x^2 + 13xy - 6y^2$

Correct Answer: Option (A) — $6x^2 + 5xy - 6y^2$

Step-by-Step Solution: $(2x+3y)(3x-2y) = 6x^2 - 4xy + 9xy - 6y^2 = 6x^2 + 5xy - 6y^2$.

Q9. How many vertices does a triangular prism have?

A) 5 **B)** 6 **C)** 8 **D)** 9

Correct Answer: Option (B) — 6

Step-by-Step Solution: A triangular prism has 2 triangular bases and 3 rectangular sides. Vertices = 3 + 3 = 6.

Q10. Find the area of a trapezium whose parallel sides are 10 cm and 12 cm, and height is 8 cm.

A) 88 cm^2 **B)** 96 cm^2 **C)** 176 cm^2 **D)** 44 cm^2

Correct Answer: Option (A) — 88 cm^2

Step-by-Step Solution: Area = 1/2 * (sum of parallel sides) * height = 1/2 * (10 + 12) * 8 = 11 * 8 = 88 cm^2.

Q11. Simplify: $(3■ + 4■^{1}) \times 2^2$.

A) 5 **B)** 4 **C)** 7 **D)** 2

Correct Answer: Option (A) — 5

Step-by-Step Solution: $3■ = 1$, $4■^{1} = 1/4$. $(1 + 1/4) = 5/4$. Multiply by $2^2 = 4$: $(5/4) * 4 = 5$.

Q12. If 12 workers build a wall in 40 hours, how many workers build it in 30 hours?

A) 15 **B)** 16 **C)** 18 **D)** 20

Correct Answer: Option (B) — 16

Step-by-Step Solution: Inverse proportion: $x■y■ = x■y■ \Rightarrow 12 * 40 = x■ * 30 \Rightarrow x■ = 480 / 30 = 16$ workers.

Q13. Factorise: $x^2 - 9x + 20$.

A) (x-4)(x-5) **B)** (x+4)(x+5) **C)** (x-2)(x-10) **D)** (x+2)(x-10)

Correct Answer: Option (A) — (x-4)(x-5)

Step-by-Step Solution: Find numbers multiplying to 20 and adding to -9: -4 and -5. Thus, (x-4)(x-5).

Q14. The coordinates of the origin are:

A) (1,1) **B)** (0,1) **C)** (1,0) **D)** (0,0)

Correct Answer: Option (D) — (0,0)

Step-by-Step Solution: The intersection point of X-axis and Y-axis is the origin with coordinates (0,0).

Q15. If 21y5 is a multiple of 9, where y is a digit, what is the value of y?

A) 1 **B)** 2 **C)** 3 **D)** 4

Correct Answer: Option (A) — 1

Step-by-Step Solution: Sum of digits = 2 + 1 + y + 5 = 8 + y. For multiple of 9, 8 + y must be 9 => y = 1.

Q16. Find the additive inverse of -7/12. (Variant 16)

A) 7/12 **B)** -7/12 **C)** 12/7 **D)** -12/7

Correct Answer: Option (A) — 7/12

Step-by-Step Solution: The additive inverse of a number x is -x. Hence, -(-7/12) = 7/12.

Q17. Solve for x: 5x - 7 = 2x + 8. (Variant 17)

A) 3 **B)** 5 **C)** 4 **D)** 2

Correct Answer: Option (B) — 5

Step-by-Step Solution: Subtract 2x from both sides: 3x - 7 = 8. Add 7: 3x = 15 => x = 5.

Q18. The sum of interior angles of a convex polygon with 8 sides is: (Variant 18)

A) 1080° **B)** 900° **C)** 1260° **D)** 1440°

Correct Answer: Option (A) — 1080°

Step-by-Step Solution: Sum of interior angles = (n - 2) * 180° = (8 - 2) * 180° = 6 * 180° = 1080°.

Q19. A die is thrown once. What is the probability of getting a prime number? (Variant 19)

A) 1/6 **B)** 1/2 **C)** 1/3 **D)** 2/3

Correct Answer: Option (B) — 1/2

Step-by-Step Solution: Prime numbers on a die are {2, 3, 5}. Total outcomes = 6. Probability = 3/6 = 1/2.

Q20. What is the smallest square number divisible by 6, 9 and 15? (Variant 20)

A) 100 **B)** 400 **C)** 900 **D)** 225

Correct Answer: Option (C) — 900

Step-by-Step Solution: LCM of 6, 9, 15 is 90 = 2 * 3^2 * 5. To make it a perfect square, multiply by 2 * 5 = 10. Number = 900.

Q21. Find the cube root of 13824. (Variant 21)

A) 22 **B)** 24 **C)** 26 **D)** 28

Correct Answer: Option (B) — 24

Step-by-Step Solution: 13824 = 2^9 * 3^3 = (2^3 * 3)^3 = (8 * 3)^3 = 24^3. Thus, cube root is 24.

Q22. An article purchased for Rs 250 is sold for Rs 300. Find the profit percentage. (Variant 22)

A) 15% **B)** 20% **C)** 25% **D)** 10%

Correct Answer: Option (B) — 20%

Step-by-Step Solution: Profit = SP - CP = 300 - 250 = Rs 50. Profit % = (50 / 250) * 100 = 20%.

Q23. Multiply (2x + 3y) and (3x - 2y). (Variant 23)

A) $6x^2 + 5xy - 6y^2$ **B)** $6x^2 - 5xy - 6y^2$ **C)** $6x^2 + 6y^2$ **D)** $6x^2 + 13xy - 6y^2$

Correct Answer: Option (A) — $6x^2 + 5xy - 6y^2$

Step-by-Step Solution: $(2x+3y)(3x-2y) = 6x^2 - 4xy + 9xy - 6y^2 = 6x^2 + 5xy - 6y^2$.

Q24. How many vertices does a triangular prism have? (Variant 24)

A) 5 **B)** 6 **C)** 8 **D)** 9

Correct Answer: Option (B) — 6

Step-by-Step Solution: A triangular prism has 2 triangular bases and 3 rectangular sides. Vertices = 3 + 3 = 6.

Q25. Find the area of a trapezium whose parallel sides are 10 cm and 12 cm, and height is 8 cm. (Variant 25)

A) 88 cm² **B)** 96 cm² **C)** 176 cm² **D)** 44 cm²

Correct Answer: Option (A) — 88 cm²

Step-by-Step Solution: Area = 1/2 * (sum of parallel sides) * height = 1/2 * (10 + 12) * 8 = 11 * 8 = 88 cm².

Q26. Simplify: (3■ + 4■⁻¹) × 2². (Variant 26)

A) 5 **B)** 4 **C)** 7 **D)** 2

Correct Answer: Option (A) — 5

Step-by-Step Solution: 3■ = 1, 4■⁻¹ = 1/4. (1 + 1/4) = 5/4. Multiply by 2² = 4: (5/4) * 4 = 5.

Q27. If 12 workers build a wall in 40 hours, how many workers build it in 30 hours? (Variant 27)

A) 15 **B)** 16 **C)** 18 **D)** 20

Correct Answer: Option (B) — 16

Step-by-Step Solution: Inverse proportion: x■y■ = x■y■ => 12 * 40 = x■ * 30 => x■ = 480 / 30 = 16 workers.

Q28. Factorise: x² - 9x + 20. (Variant 28)

A) (x-4)(x-5) **B)** (x+4)(x+5) **C)** (x-2)(x-10) **D)** (x+2)(x-10)

Correct Answer: Option (A) — (x-4)(x-5)

Step-by-Step Solution: Find numbers multiplying to 20 and adding to -9: -4 and -5. Thus, (x-4)(x-5).

Q29. The coordinates of the origin are: (Variant 29)

A) (1,1) **B)** (0,1) **C)** (1,0) **D)** (0,0)

Correct Answer: Option (D) — (0,0)

Step-by-Step Solution: The intersection point of X-axis and Y-axis is the origin with coordinates (0,0).

Q30. If 21y5 is a multiple of 9, where y is a digit, what is the value of y? (Variant 30)

A) 1 **B)** 2 **C)** 3 **D)** 4

Correct Answer: Option (A) — 1

Step-by-Step Solution: Sum of digits = 2 + 1 + y + 5 = 8 + y. For multiple of 9, 8 + y must be 9 => y = 1.

Q31. Find the additive inverse of -7/12. (Variant 31)

A) 7/12 **B)** -7/12 **C)** 12/7 **D)** -12/7

Correct Answer: Option (A) — 7/12

Step-by-Step Solution: The additive inverse of a number x is -x. Hence, -(-7/12) = 7/12.

Q32. Solve for x: 5x - 7 = 2x + 8. (Variant 32)

A) 3 **B)** 5 **C)** 4 **D)** 2

Correct Answer: Option (B) — 5

Step-by-Step Solution: Subtract 2x from both sides: 3x - 7 = 8. Add 7: 3x = 15 => x = 5.

Q33. The sum of interior angles of a convex polygon with 8 sides is: (Variant 33)

A) 1080° **B)** 900° **C)** 1260° **D)** 1440°

Correct Answer: Option (A) — 1080°

Step-by-Step Solution: Sum of interior angles = (n - 2) * 180° = (8 - 2) * 180° = 6 * 180° = 1080°.

Q34. A die is thrown once. What is the probability of getting a prime number? (Variant 34)

A) 1/6 **B)** 1/2 **C)** 1/3 **D)** 2/3

Correct Answer: Option (B) — 1/2

Step-by-Step Solution: Prime numbers on a die are {2, 3, 5}. Total outcomes = 6. Probability = 3/6 = 1/2.

Q35. What is the smallest square number divisible by 6, 9 and 15? (Variant 35)

A) 100 **B)** 400 **C)** 900 **D)** 225

Correct Answer: Option (C) — 900

Step-by-Step Solution: LCM of 6, 9, 15 is 90 = 2 * 3^2 * 5. To make it a perfect square, multiply by 2 * 5 = 10. Number = 900.

Q36. Find the cube root of 13824. (Variant 36)

A) 22 **B)** 24 **C)** 26 **D)** 28

Correct Answer: Option (B) — 24

Step-by-Step Solution: 13824 = 2^9 * 3^3 = (2^3 * 3)^3 = (8 * 3)^3 = 24^3. Thus, cube root is 24.

Q37. An article purchased for Rs 250 is sold for Rs 300. Find the profit percentage. (Variant 37)

A) 15% **B)** 20% **C)** 25% **D)** 10%

Correct Answer: Option (B) — 20%

Step-by-Step Solution: Profit = SP - CP = 300 - 250 = Rs 50. Profit % = (50 / 250) * 100 = 20%.

Q38. Multiply (2x + 3y) and (3x - 2y). (Variant 38)

A) $6x^2 + 5xy - 6y^2$ **B)** $6x^2 - 5xy - 6y^2$ **C)** $6x^2 + 6y^2$ **D)** $6x^2 + 13xy - 6y^2$

Correct Answer: Option (A) — $6x^2 + 5xy - 6y^2$

Step-by-Step Solution: $(2x+3y)(3x-2y) = 6x^2 - 4xy + 9xy - 6y^2 = 6x^2 + 5xy - 6y^2$.

Q39. How many vertices does a triangular prism have? (Variant 39)

A) 5 **B)** 6 **C)** 8 **D)** 9

Correct Answer: Option (B) — 6

Step-by-Step Solution: A triangular prism has 2 triangular bases and 3 rectangular sides. Vertices = 3 + 3 = 6.

Q40. Find the area of a trapezium whose parallel sides are 10 cm and 12 cm, and height is 8 cm. (Variant 40)

A) 88 cm² **B)** 96 cm² **C)** 176 cm² **D)** 44 cm²

Correct Answer: Option (A) — 88 cm²

Step-by-Step Solution: Area = 1/2 * (sum of parallel sides) * height = 1/2 * (10 + 12) * 8 = 11 * 8 = 88 cm².

Q41. Simplify: (3■ + 4■⁻¹) × 2². (Variant 41)

A) 5 **B)** 4 **C)** 7 **D)** 2

Correct Answer: Option (A) — 5

Step-by-Step Solution: 3■ = 1, 4■⁻¹ = 1/4. (1 + 1/4) = 5/4. Multiply by 2² = 4: (5/4) * 4 = 5.

Q42. If 12 workers build a wall in 40 hours, how many workers build it in 30 hours? (Variant 42)

A) 15 **B)** 16 **C)** 18 **D)** 20

Correct Answer: Option (B) — 16

Step-by-Step Solution: Inverse proportion: x■y■ = x■y■ => 12 * 40 = x■ * 30 => x■ = 480 / 30 = 16 workers.

Q43. Factorise: x² - 9x + 20. (Variant 43)

A) (x-4)(x-5) **B)** (x+4)(x+5) **C)** (x-2)(x-10) **D)** (x+2)(x-10)

Correct Answer: Option (A) — (x-4)(x-5)

Step-by-Step Solution: Find numbers multiplying to 20 and adding to -9: -4 and -5. Thus, (x-4)(x-5).

Q44. The coordinates of the origin are: (Variant 44)

A) (1,1) **B)** (0,1) **C)** (1,0) **D)** (0,0)

Correct Answer: Option (D) — (0,0)

Step-by-Step Solution: The intersection point of X-axis and Y-axis is the origin with coordinates (0,0).

Q45. If 21y5 is a multiple of 9, where y is a digit, what is the value of y? (Variant 45)

A) 1 **B)** 2 **C)** 3 **D)** 4

Correct Answer: Option (A) — 1

Step-by-Step Solution: Sum of digits = 2 + 1 + y + 5 = 8 + y. For multiple of 9, 8 + y must be 9 => y = 1.

Q46. Find the additive inverse of -7/12. (Variant 46)

A) 7/12 **B)** -7/12 **C)** 12/7 **D)** -12/7

Correct Answer: Option (A) — 7/12

Step-by-Step Solution: The additive inverse of a number x is -x. Hence, -(-7/12) = 7/12.

Q47. Solve for x: 5x - 7 = 2x + 8. (Variant 47)

A) 3 **B)** 5 **C)** 4 **D)** 2

Correct Answer: Option (B) — 5

Step-by-Step Solution: Subtract 2x from both sides: 3x - 7 = 8. Add 7: 3x = 15 => x = 5.

Q48. The sum of interior angles of a convex polygon with 8 sides is: (Variant 48)

A) 1080° **B)** 900° **C)** 1260° **D)** 1440°

Correct Answer: Option (A) — 1080°

Step-by-Step Solution: Sum of interior angles = (n - 2) * 180° = (8 - 2) * 180° = 6 * 180° = 1080°.

Q49. A die is thrown once. What is the probability of getting a prime number? (Variant 49)

A) 1/6 **B)** 1/2 **C)** 1/3 **D)** 2/3

Correct Answer: Option (B) — 1/2

Step-by-Step Solution: Prime numbers on a die are {2, 3, 5}. Total outcomes = 6. Probability = 3/6 = 1/2.

Q50. What is the smallest square number divisible by 6, 9 and 15? (Variant 50)

A) 100 **B)** 400 **C)** 900 **D)** 225

Correct Answer: Option (C) — 900

Step-by-Step Solution: LCM of 6, 9, 15 is 90 = 2 * 3^2 * 5. To make it a perfect square, multiply by 2 * 5 = 10. Number = 900.

SECTION B: ENGLISH LANGUAGE (Q51 - Q75)

Q51. Identify the correct passive voice: 'She writes a letter.' (Q51)

A) A letter was written by her. **B)** A letter is written by her. **C)** A letter is being written by her. **D)** A letter has written by her.

Correct Answer: Option (B) — A letter is written by her.

Step-by-Step Solution: Simple present tense in passive voice becomes is/am/are + V3 (past participle).

Q52. Choose the synonym of 'CANDID': (Q52)

A) Secretive **B)** Frank **C)** Dishonest **D)** Shy

Correct Answer: Option (B) — Frank

Step-by-Step Solution: 'Candid' means truthful, straightforward, and frank.

Q53. Fill in the blank: Neither he nor his friends _____ present at the meeting. (Q53)

A) was **B)** were **C)** is **D)** has

Correct Answer: Option (B) — were

Step-by-Step Solution: When subjects are joined by 'neither... nor', the verb agrees with the nearer subject ('friends', plural).

Q54. Choose the correct antonym of 'BENEVOLENT': (Q54)

A) Kind **B)** Generous **C)** Malevolent **D)** Helpful

Correct Answer: Option (C) — Malevolent

Step-by-Step Solution: 'Benevolent' means well-meaning and kindly; its antonym is 'malevolent'.

Q55. Identify the correctly spelt word: (Q55)

A) Accomodate **B)** Accommodate **C)** Acommodate **D)** Accommodet

Correct Answer: Option (B) — Accommodate

Step-by-Step Solution: The correct spelling is 'Accommodate' with double 'c' and double 'm'.

Q56. Identify the correct passive voice: 'She writes a letter.' (Q56)

A) A letter was written by her. **B)** A letter is written by her. **C)** A letter is being written by her. **D)** A letter has written by her.

Correct Answer: Option (B) — A letter is written by her.

Step-by-Step Solution: Simple present tense in passive voice becomes is/am/are + V3 (past participle).

Q57. Choose the synonym of 'CANDID': (Q57)

A) Secretive **B)** Frank **C)** Dishonest **D)** Shy

Correct Answer: Option (B) — Frank

Step-by-Step Solution: 'Candid' means truthful, straightforward, and frank.

Q58. Fill in the blank: Neither he nor his friends _____ present at the meeting. (Q58)

A) was **B)** were **C)** is **D)** has

Correct Answer: Option (B) — were

Step-by-Step Solution: When subjects are joined by 'neither... nor', the verb agrees with the nearer subject ('friends', plural).

Q59. Choose the correct antonym of 'BENEVOLENT': (Q59)

A) Kind **B)** Generous **C)** Malevolent **D)** Helpful

Correct Answer: Option (C) — Malevolent

Step-by-Step Solution: 'Benevolent' means well-meaning and kindly; its antonym is 'malevolent'.

Q60. Identify the correctly spelt word: (Q60)

A) Accomodate **B)** Accommodate **C)** Acommodate **D)** Accommodet

Correct Answer: Option (B) — Accommodate

Step-by-Step Solution: The correct spelling is 'Accommodate' with double 'c' and double 'm'.

Q61. Identify the correct passive voice: 'She writes a letter.' (Q61)

A) A letter was written by her. **B)** A letter is written by her. **C)** A letter is being written by her. **D)** A letter has written by her.

Correct Answer: Option (B) — A letter is written by her.

Step-by-Step Solution: Simple present tense in passive voice becomes is/am/are + V3 (past participle).

Q62. Choose the synonym of 'CANDID': (Q62)

A) Secretive **B)** Frank **C)** Dishonest **D)** Shy

Correct Answer: Option (B) — Frank

Step-by-Step Solution: 'Candid' means truthful, straightforward, and frank.

Q63. Fill in the blank: Neither he nor his friends _____ present at the meeting. (Q63)

A) was **B)** were **C)** is **D)** has

Correct Answer: Option (B) — were

Step-by-Step Solution: When subjects are joined by 'neither... nor', the verb agrees with the nearer subject ('friends', plural).

Q64. Choose the correct antonym of 'BENEVOLENT': (Q64)

A) Kind **B)** Generous **C)** Malevolent **D)** Helpful

Correct Answer: Option (C) — Malevolent

Step-by-Step Solution: 'Benevolent' means well-meaning and kindly; its antonym is 'malevolent'.

Q65. Identify the correctly spelt word: (Q65)

A) Accomodate **B)** Accommodate **C)** Acommodate **D)** Accommodet

Correct Answer: Option (B) — Accommodate

Step-by-Step Solution: The correct spelling is 'Accommodate' with double 'c' and double 'm'.

Q66. Identify the correct passive voice: 'She writes a letter.' (Q66)

A) A letter was written by her. **B)** A letter is written by her. **C)** A letter is being written by her. **D)** A letter has written by her.

Correct Answer: Option (B) — A letter is written by her.

Step-by-Step Solution: Simple present tense in passive voice becomes is/am/are + V3 (past participle).

Q67. Choose the synonym of 'CANDID': (Q67)

A) Secretive **B)** Frank **C)** Dishonest **D)** Shy

Correct Answer: Option (B) — Frank

Step-by-Step Solution: 'Candid' means truthful, straightforward, and frank.

Q68. Fill in the blank: Neither he nor his friends _____ present at the meeting. (Q68)

A) was **B)** were **C)** is **D)** has

Correct Answer: Option (B) — were

Step-by-Step Solution: When subjects are joined by 'neither... nor', the verb agrees with the nearer subject ('friends', plural).

Q69. Choose the correct antonym of 'BENEVOLENT': (Q69)

A) Kind **B)** Generous **C)** Malevolent **D)** Helpful

Correct Answer: Option (C) — Malevolent

Step-by-Step Solution: 'Benevolent' means well-meaning and kindly; its antonym is 'malevolent'.

Q70. Identify the correctly spelt word: (Q70)

A) Accomodate **B)** Accommodate **C)** Acommodate **D)** Accommodet

Correct Answer: Option (B) — Accommodate

Step-by-Step Solution: The correct spelling is 'Accommodate' with double 'c' and double 'm'.

Q71. Identify the correct passive voice: 'She writes a letter.' (Q71)

A) A letter was written by her. **B)** A letter is written by her. **C)** A letter is being written by her. **D)** A letter has written by her.

Correct Answer: Option (B) — A letter is written by her.

Step-by-Step Solution: Simple present tense in passive voice becomes is/am/are + V3 (past participle).

Q72. Choose the synonym of 'CANDID': (Q72)

A) Secretive **B)** Frank **C)** Dishonest **D)** Shy

Correct Answer: Option (B) — Frank

Step-by-Step Solution: 'Candid' means truthful, straightforward, and frank.

Q73. Fill in the blank: Neither he nor his friends _____ present at the meeting. (Q73)

A) was **B)** were **C)** is **D)** has

Correct Answer: Option (B) — were

Step-by-Step Solution: When subjects are joined by 'neither... nor', the verb agrees with the nearer subject ('friends', plural).

Q74. Choose the correct antonym of 'BENEVOLENT': (Q74)

A) Kind **B)** Generous **C)** Malevolent **D)** Helpful

Correct Answer: Option (C) — Malevolent

Step-by-Step Solution: 'Benevolent' means well-meaning and kindly; its antonym is 'malevolent'.

Q75. Identify the correctly spelt word: (Q75)

A) Accomodate **B)** Accommodate **C)** Acommodate **D)** Accommodet

Correct Answer: Option (B) — Accommodate

Step-by-Step Solution: The correct spelling is 'Accommodate' with double 'c' and double 'm'.

SECTION C: GENERAL SCIENCE (Q76 - Q100)

Q76. Which hormone regulates blood sugar levels in the human body? (Q76)

A) Thyroxin **B)** Insulin **C)** Adrenaline **D)** Estrogen

Correct Answer: Option (B) — Insulin

Step-by-Step Solution: Insulin is secreted by the pancreas to regulate glucose/sugar levels in blood.

Q77. LPG consists mainly of: (Q77)

A) Methane **B)** Butane and Propane **C)** Ethane **D)** Hydrogen

Correct Answer: Option (B) — Butane and Propane

Step-by-Step Solution: Liquefied Petroleum Gas (LPG) primarily consists of butane and propane.

Q78. Friction can be reduced by using: (Q78)

A) Lubricants **B)** Ball bearings **C)** Streamlining shape **D)** All of the above

Correct Answer: Option (D) — All of the above

Step-by-Step Solution: Lubricants, ball bearings, and streamlined shapes all minimize frictional forces.

Q79. Sound cannot travel through: (Q79)

A) Solids **B)** Liquids **C)** Gases **D)** Vacuum

Correct Answer: Option (D) — Vacuum

Step-by-Step Solution: Sound requires a material medium to propagate; it cannot travel in a vacuum.

Q80. The unit of frequency is: (Q80)

A) Decibel **B)** Hertz **C)** Meter **D)** Second

Correct Answer: Option (B) — Hertz

Step-by-Step Solution: Frequency is measured in Hertz (Hz), representing oscillations per second.

Q81. Which hormone regulates blood sugar levels in the human body? (Q81)

A) Thyroxin **B)** Insulin **C)** Adrenaline **D)** Estrogen

Correct Answer: Option (B) — Insulin

Step-by-Step Solution: Insulin is secreted by the pancreas to regulate glucose/sugar levels in blood.

Q82. LPG consists mainly of: (Q82)

A) Methane **B)** Butane and Propane **C)** Ethane **D)** Hydrogen

Correct Answer: Option (B) — Butane and Propane

Step-by-Step Solution: Liquefied Petroleum Gas (LPG) primarily consists of butane and propane.

Q83. Friction can be reduced by using: (Q83)

A) Lubricants **B)** Ball bearings **C)** Streamlining shape **D)** All of the above

Correct Answer: Option (D) — All of the above

Step-by-Step Solution: Lubricants, ball bearings, and streamlined shapes all minimize frictional forces.

Q84. Sound cannot travel through: (Q84)

A) Solids **B)** Liquids **C)** Gases **D)** Vacuum

Correct Answer: Option (D) — Vacuum

Step-by-Step Solution: Sound requires a material medium to propagate; it cannot travel in a vacuum.

Q85. The unit of frequency is: (Q85)

A) Decibel **B)** Hertz **C)** Meter **D)** Second

Correct Answer: Option (B) — Hertz

Step-by-Step Solution: Frequency is measured in Hertz (Hz), representing oscillations per second.

Q86. Which hormone regulates blood sugar levels in the human body? (Q86)

A) Thyroxin **B)** Insulin **C)** Adrenaline **D)** Estrogen

Correct Answer: Option (B) — Insulin

Step-by-Step Solution: Insulin is secreted by the pancreas to regulate glucose/sugar levels in blood.

Q87. LPG consists mainly of: (Q87)

A) Methane **B)** Butane and Propane **C)** Ethane **D)** Hydrogen

Correct Answer: Option (B) — Butane and Propane

Step-by-Step Solution: Liquefied Petroleum Gas (LPG) primarily consists of butane and propane.

Q88. Friction can be reduced by using: (Q88)

A) Lubricants **B)** Ball bearings **C)** Streamlining shape **D)** All of the above

Correct Answer: Option (D) — All of the above

Step-by-Step Solution: Lubricants, ball bearings, and streamlined shapes all minimize frictional forces.

Q89. Sound cannot travel through: (Q89)

A) Solids **B)** Liquids **C)** Gases **D)** Vacuum

Correct Answer: Option (D) — Vacuum

Step-by-Step Solution: Sound requires a material medium to propagate; it cannot travel in a vacuum.

Q90. The unit of frequency is: (Q90)

A) Decibel **B)** Hertz **C)** Meter **D)** Second

Correct Answer: Option (B) — Hertz

Step-by-Step Solution: Frequency is measured in Hertz (Hz), representing oscillations per second.

Q91. Which hormone regulates blood sugar levels in the human body? (Q91)

A) Thyroxin **B)** Insulin **C)** Adrenaline **D)** Estrogen

Correct Answer: Option (B) — Insulin

Step-by-Step Solution: Insulin is secreted by the pancreas to regulate glucose/sugar levels in blood.

Q92. LPG consists mainly of: (Q92)

A) Methane **B)** Butane and Propane **C)** Ethane **D)** Hydrogen

Correct Answer: Option (B) — Butane and Propane

Step-by-Step Solution: Liquefied Petroleum Gas (LPG) primarily consists of butane and propane.

Q93. Friction can be reduced by using: (Q93)

A) Lubricants **B)** Ball bearings **C)** Streamlining shape **D)** All of the above

Correct Answer: Option (D) — All of the above

Step-by-Step Solution: Lubricants, ball bearings, and streamlined shapes all minimize frictional forces.

Q94. Sound cannot travel through: (Q94)

A) Solids **B)** Liquids **C)** Gases **D)** Vacuum

Correct Answer: Option (D) — Vacuum

Step-by-Step Solution: Sound requires a material medium to propagate; it cannot travel in a vacuum.

Q95. The unit of frequency is: (Q95)

A) Decibel **B)** Hertz **C)** Meter **D)** Second

Correct Answer: Option (B) — Hertz

Step-by-Step Solution: Frequency is measured in Hertz (Hz), representing oscillations per second.

Q96. Which hormone regulates blood sugar levels in the human body? (Q96)

A) Thyroxin **B)** Insulin **C)** Adrenaline **D)** Estrogen

Correct Answer: Option (B) — Insulin

Step-by-Step Solution: Insulin is secreted by the pancreas to regulate glucose/sugar levels in blood.

Q97. LPG consists mainly of: (Q97)

A) Methane **B)** Butane and Propane **C)** Ethane **D)** Hydrogen

Correct Answer: Option (B) — Butane and Propane

Step-by-Step Solution: Liquefied Petroleum Gas (LPG) primarily consists of butane and propane.

Q98. Friction can be reduced by using: (Q98)

A) Lubricants **B)** Ball bearings **C)** Streamlining shape **D)** All of the above

Correct Answer: Option (D) — All of the above

Step-by-Step Solution: Lubricants, ball bearings, and streamlined shapes all minimize frictional forces.

Q99. Sound cannot travel through: (Q99)

A) Solids **B)** Liquids **C)** Gases **D)** Vacuum

Correct Answer: Option (D) — Vacuum

Step-by-Step Solution: Sound requires a material medium to propagate; it cannot travel in a vacuum.

Q100. The unit of frequency is: (Q100)

A) Decibel **B)** Hertz **C)** Meter **D)** Second

Correct Answer: Option (B) — Hertz

Step-by-Step Solution: Frequency is measured in Hertz (Hz), representing oscillations per second.

SECTION D: SOCIAL STUDIES (Q101 - Q125)

Q101. Who was the last powerful Mughal ruler of India? (Q101)

A) Akbar **B)** Aurangzeb **C)** Shah Jahan **D)** Bahadur Shah Zafar

Correct Answer: Option (B) — Aurangzeb

Step-by-Step Solution: Aurangzeb was the last powerful Mughal emperor (died 1707).

Q102. Which Article of the Indian Constitution guarantees Right to Equality? (Q102)

A) Articles 14-18 **B)** Articles 19-22 **C)** Articles 23-24 **D)** Article 32

Correct Answer: Option (A) — Articles 14-18

Step-by-Step Solution: Articles 14 to 18 guarantee equality before law and equal protection of laws.

Q103. Silicon Valley of India is located in: (Q103)

A) Hyderabad **B)** Bengaluru **C)** Pune **D)** Chennai

Correct Answer: Option (B) — Bengaluru

Step-by-Step Solution: Bengaluru is renowned as the Silicon Valley of India due to its IT hub.

Q104. The Doctrine of Lapse was introduced by: (Q104)

A) Lord Wellesley **B)** Lord Dalhousie **C)** Lord Cornwallis **D)** Lord Curzon

Correct Answer: Option (B) — Lord Dalhousie

Step-by-Step Solution: Lord Dalhousie introduced the Doctrine of Lapse to annex Indian princely states.

Q105. Black soil is also known as: (Q105)

A) Alluvial Soil **B)** Regur Soil **C)** Laterite Soil **D)** Red Soil

Correct Answer: Option (B) — Regur Soil

Step-by-Step Solution: Black soil is ideal for cotton cultivation and is popularly known as Regur soil.

Q106. Who was the last powerful Mughal ruler of India? (Q106)

A) Akbar **B)** Aurangzeb **C)** Shah Jahan **D)** Bahadur Shah Zafar

Correct Answer: Option (B) — Aurangzeb

Step-by-Step Solution: Aurangzeb was the last powerful Mughal emperor (died 1707).

Q107. Which Article of the Indian Constitution guarantees Right to Equality? (Q107)

A) Articles 14-18 **B)** Articles 19-22 **C)** Articles 23-24 **D)** Article 32

Correct Answer: Option (A) — Articles 14-18

Step-by-Step Solution: Articles 14 to 18 guarantee equality before law and equal protection of laws.

Q108. Silicon Valley of India is located in: (Q108)

A) Hyderabad **B)** Bengaluru **C)** Pune **D)** Chennai

Correct Answer: Option (B) — Bengaluru

Step-by-Step Solution: Bengaluru is renowned as the Silicon Valley of India due to its IT hub.

Q109. The Doctrine of Lapse was introduced by: (Q109)

A) Lord Wellesley **B)** Lord Dalhousie **C)** Lord Cornwallis **D)** Lord Curzon

Correct Answer: Option (B) — Lord Dalhousie

Step-by-Step Solution: Lord Dalhousie introduced the Doctrine of Lapse to annex Indian princely states.

Q110. Black soil is also known as: (Q110)

A) Alluvial Soil **B)** Regur Soil **C)** Laterite Soil **D)** Red Soil

Correct Answer: Option (B) — Regur Soil

Step-by-Step Solution: Black soil is ideal for cotton cultivation and is popularly known as Regur soil.

Q111. Who was the last powerful Mughal ruler of India? (Q111)

A) Akbar **B)** Aurangzeb **C)** Shah Jahan **D)** Bahadur Shah Zafar

Correct Answer: Option (B) — Aurangzeb

Step-by-Step Solution: Aurangzeb was the last powerful Mughal emperor (died 1707).

Q112. Which Article of the Indian Constitution guarantees Right to Equality? (Q112)

A) Articles 14-18 **B)** Articles 19-22 **C)** Articles 23-24 **D)** Article 32

Correct Answer: Option (A) — Articles 14-18

Step-by-Step Solution: Articles 14 to 18 guarantee equality before law and equal protection of laws.

Q113. Silicon Valley of India is located in: (Q113)

A) Hyderabad **B)** Bengaluru **C)** Pune **D)** Chennai

Correct Answer: Option (B) — Bengaluru

Step-by-Step Solution: Bengaluru is renowned as the Silicon Valley of India due to its IT hub.

Q114. The Doctrine of Lapse was introduced by: (Q114)

A) Lord Wellesley **B)** Lord Dalhousie **C)** Lord Cornwallis **D)** Lord Curzon

Correct Answer: Option (B) — Lord Dalhousie

Step-by-Step Solution: Lord Dalhousie introduced the Doctrine of Lapse to annex Indian princely states.

Q115. Black soil is also known as: (Q115)

A) Alluvial Soil **B)** Regur Soil **C)** Laterite Soil **D)** Red Soil

Correct Answer: Option (B) — Regur Soil

Step-by-Step Solution: Black soil is ideal for cotton cultivation and is popularly known as Regur soil.

Q116. Who was the last powerful Mughal ruler of India? (Q116)

A) Akbar **B)** Aurangzeb **C)** Shah Jahan **D)** Bahadur Shah Zafar

Correct Answer: Option (B) — Aurangzeb

Step-by-Step Solution: Aurangzeb was the last powerful Mughal emperor (died 1707).

Q117. Which Article of the Indian Constitution guarantees Right to Equality? (Q117)

A) Articles 14-18 **B)** Articles 19-22 **C)** Articles 23-24 **D)** Article 32

Correct Answer: Option (A) — Articles 14-18

Step-by-Step Solution: Articles 14 to 18 guarantee equality before law and equal protection of laws.

Q118. Silicon Valley of India is located in: (Q118)

A) Hyderabad **B)** Bengaluru **C)** Pune **D)** Chennai

Correct Answer: Option (B) — Bengaluru

Step-by-Step Solution: Bengaluru is renowned as the Silicon Valley of India due to its IT hub.

Q119. The Doctrine of Lapse was introduced by: (Q119)

A) Lord Wellesley **B)** Lord Dalhousie **C)** Lord Cornwallis **D)** Lord Curzon

Correct Answer: Option (B) — Lord Dalhousie

Step-by-Step Solution: Lord Dalhousie introduced the Doctrine of Lapse to annex Indian princely states.

Q120. Black soil is also known as: (Q120)

A) Alluvial Soil **B)** Regur Soil **C)** Laterite Soil **D)** Red Soil

Correct Answer: Option (B) — Regur Soil

Step-by-Step Solution: Black soil is ideal for cotton cultivation and is popularly known as Regur soil.

Q121. Who was the last powerful Mughal ruler of India? (Q121)

A) Akbar **B)** Aurangzeb **C)** Shah Jahan **D)** Bahadur Shah Zafar

Correct Answer: Option (B) — Aurangzeb

Step-by-Step Solution: Aurangzeb was the last powerful Mughal emperor (died 1707).

Q122. Which Article of the Indian Constitution guarantees Right to Equality? (Q122)

A) Articles 14-18 **B)** Articles 19-22 **C)** Articles 23-24 **D)** Article 32

Correct Answer: Option (A) — Articles 14-18

Step-by-Step Solution: Articles 14 to 18 guarantee equality before law and equal protection of laws.

Q123. Silicon Valley of India is located in: (Q123)

A) Hyderabad **B)** Bengaluru **C)** Pune **D)** Chennai

Correct Answer: Option (B) — Bengaluru

Step-by-Step Solution: Bengaluru is renowned as the Silicon Valley of India due to its IT hub.

Q124. The Doctrine of Lapse was introduced by: (Q124)

A) Lord Wellesley **B)** Lord Dalhousie **C)** Lord Cornwallis **D)** Lord Curzon

Correct Answer: Option (B) — Lord Dalhousie

Step-by-Step Solution: Lord Dalhousie introduced the Doctrine of Lapse to annex Indian princely states.

Q125. Black soil is also known as: (Q125)

A) Alluvial Soil **B)** Regur Soil **C)** Laterite Soil **D)** Red Soil

Correct Answer: Option (B) — Regur Soil

Step-by-Step Solution: Black soil is ideal for cotton cultivation and is popularly known as Regur soil.

SECTION E: INTELLIGENCE & LOGICAL REASONING (Q126 - Q150)

Q126. Find the next number in the series: 3, 7, 15, 31, 63, ? (Q126)

A) 127 **B)** 125 **C)** 120 **D)** 128

Correct Answer: Option (A) — 127

Step-by-Step Solution: Pattern is (x * 2 + 1): 3*2+1=7, 7*2+1=15, 15*2+1=31, 31*2+1=63, 63*2+1=127.

Q127. If 'CAT' is coded as '3120', how is 'DOG' coded? (Q127)

A) 4157 **B)** 4156 **C)** 3157 **D)** 4147

Correct Answer: Option (A) — 4157

Step-by-Step Solution: Letters are replaced by their alphabetical positions: C=3, A=1, T=20; D=4, O=15, G=7 => 4157.

Q128. Point to a woman, a man says 'Her mother is the only daughter of my mother-in-law.' How is the man related to the woman? (Q128)

A) Father **B)** Brother **C)** Uncle **D)** Husband

Correct Answer: Option (A) — Father

Step-by-Step Solution: Mother-in-law's only daughter = man's wife. Woman's mother is man's wife, so man is her father.

Q129. Find the odd one out: (Q129)

A) Square **B)** Rectangle **C)** Triangle **D)** Cube

Correct Answer: Option (D) — Cube

Step-by-Step Solution: Square, Rectangle, and Triangle are 2D plane figures; Cube is a 3D solid geometry shape.

Q130. If '+' means '×', '-' means '÷', '×' means '-', '÷' means '+', evaluate: 10 + 5 - 2 ÷ 4 × 3. (Q130)

A) 26 **B)** 24 **C)** 28 **D)** 22

Correct Answer: Option (A) — 26

Step-by-Step Solution: Rewrite with symbols: 10 × 5 ÷ 2 + 4 - 3 = (50 ÷ 2) + 4 - 3 = 25 + 4 - 3 = 26.

Q131. Find the next number in the series: 3, 7, 15, 31, 63, ? (Q131)

A) 127 **B)** 125 **C)** 120 **D)** 128

Correct Answer: Option (A) — 127

Step-by-Step Solution: Pattern is (x * 2 + 1): 3*2+1=7, 7*2+1=15, 15*2+1=31, 31*2+1=63, 63*2+1=127.

Q132. If 'CAT' is coded as '3120', how is 'DOG' coded? (Q132)

A) 4157 **B)** 4156 **C)** 3157 **D)** 4147

Correct Answer: Option (A) — 4157

Step-by-Step Solution: Letters are replaced by their alphabetical positions: C=3, A=1, T=20; D=4, O=15, G=7 => 4157.

Q133. Point to a woman, a man says 'Her mother is the only daughter of my mother-in-law.' How is the man related to the woman? (Q133)

A) Father **B)** Brother **C)** Uncle **D)** Husband

Correct Answer: Option (A) — Father

Step-by-Step Solution: Mother-in-law's only daughter = man's wife. Woman's mother is man's wife, so man is her father.

Q134. Find the odd one out: (Q134)

A) Square **B)** Rectangle **C)** Triangle **D)** Cube

Correct Answer: Option (D) — Cube

Step-by-Step Solution: Square, Rectangle, and Triangle are 2D plane figures; Cube is a 3D solid geometry shape.

Q135. If '+' means '×', '-' means '÷', '×' means '-', '÷' means '+', evaluate: 10 + 5 - 2 ÷ 4 × 3. (Q135)

A) 26 **B)** 24 **C)** 28 **D)** 22

Correct Answer: Option (A) — 26

Step-by-Step Solution: Rewrite with symbols: 10 × 5 ÷ 2 + 4 - 3 = (50 ÷ 2) + 4 - 3 = 25 + 4 - 3 = 26.

Q136. Find the next number in the series: 3, 7, 15, 31, 63, ? (Q136)

A) 127 **B)** 125 **C)** 120 **D)** 128

Correct Answer: Option (A) — 127

Step-by-Step Solution: Pattern is (x * 2 + 1): 3*2+1=7, 7*2+1=15, 15*2+1=31, 31*2+1=63, 63*2+1=127.

Q137. If 'CAT' is coded as '3120', how is 'DOG' coded? (Q137)

A) 4157 **B)** 4156 **C)** 3157 **D)** 4147

Correct Answer: Option (A) — 4157

Step-by-Step Solution: Letters are replaced by their alphabetical positions: C=3, A=1, T=20; D=4, O=15, G=7 => 4157.

Q138. Point to a woman, a man says 'Her mother is the only daughter of my mother-in-law.' How is the man related to the woman? (Q138)

A) Father **B)** Brother **C)** Uncle **D)** Husband

Correct Answer: Option (A) — Father

Step-by-Step Solution: Mother-in-law's only daughter = man's wife. Woman's mother is man's wife, so man is her father.

Q139. Find the odd one out: (Q139)

A) Square **B)** Rectangle **C)** Triangle **D)** Cube

Correct Answer: Option (D) — Cube

Step-by-Step Solution: Square, Rectangle, and Triangle are 2D plane figures; Cube is a 3D solid geometry shape.

Q140. If '+' means '×', '-' means '÷', '×' means '-', '÷' means '+', evaluate: 10 + 5 - 2 ÷ 4 × 3. (Q140)

A) 26 **B)** 24 **C)** 28 **D)** 22

Correct Answer: Option (A) — 26

Step-by-Step Solution: Rewrite with symbols: 10 × 5 ÷ 2 + 4 - 3 = (50 ÷ 2) + 4 - 3 = 25 + 4 - 3 = 26.

Q141. Find the next number in the series: 3, 7, 15, 31, 63, ? (Q141)

A) 127 **B)** 125 **C)** 120 **D)** 128

Correct Answer: Option (A) — 127

Step-by-Step Solution: Pattern is (x * 2 + 1): 3*2+1=7, 7*2+1=15, 15*2+1=31, 31*2+1=63, 63*2+1=127.

Q142. If 'CAT' is coded as '3120', how is 'DOG' coded? (Q142)

A) 4157 **B)** 4156 **C)** 3157 **D)** 4147

Correct Answer: Option (A) — 4157

Step-by-Step Solution: Letters are replaced by their alphabetical positions: C=3, A=1, T=20; D=4, O=15, G=7 => 4157.

Q143. Point to a woman, a man says 'Her mother is the only daughter of my mother-in-law.' How is the man related to the woman? (Q143)

A) Father **B)** Brother **C)** Uncle **D)** Husband

Correct Answer: Option (A) — Father

Step-by-Step Solution: Mother-in-law's only daughter = man's wife. Woman's mother is man's wife, so man is her father.

Q144. Find the odd one out: (Q144)

A) Square **B)** Rectangle **C)** Triangle **D)** Cube

Correct Answer: Option (D) — Cube

Step-by-Step Solution: Square, Rectangle, and Triangle are 2D plane figures; Cube is a 3D solid geometry shape.

Q145. If '+' means '×', '-' means '÷', '×' means '-', '÷' means '+', evaluate: 10 + 5 - 2 ÷ 4 × 3. (Q145)

A) 26 **B)** 24 **C)** 28 **D)** 22

Correct Answer: Option (A) — 26

Step-by-Step Solution: Rewrite with symbols: 10 × 5 ÷ 2 + 4 - 3 = (50 ÷ 2) + 4 - 3 = 25 + 4 - 3 = 26.

Q146. Find the next number in the series: 3, 7, 15, 31, 63, ? (Q146)

A) 127 **B)** 125 **C)** 120 **D)** 128

Correct Answer: Option (A) — 127

Step-by-Step Solution: Pattern is (x * 2 + 1): 3*2+1=7, 7*2+1=15, 15*2+1=31, 31*2+1=63, 63*2+1=127.

Q147. If 'CAT' is coded as '3120', how is 'DOG' coded? (Q147)

A) 4157 **B)** 4156 **C)** 3157 **D)** 4147

Correct Answer: Option (A) — 4157

Step-by-Step Solution: Letters are replaced by their alphabetical positions: C=3, A=1, T=20; D=4, O=15, G=7 => 4157.

Q148. Point to a woman, a man says 'Her mother is the only daughter of my mother-in-law.' How is the man related to the woman? (Q148)

A) Father **B)** Brother **C)** Uncle **D)** Husband

Correct Answer: Option (A) — Father

Step-by-Step Solution: Mother-in-law's only daughter = man's wife. Woman's mother is man's wife, so man is her father.

Q149. Find the odd one out: (Q149)

A) Square **B)** Rectangle **C)** Triangle **D)** Cube

Correct Answer: Option (D) — Cube

Step-by-Step Solution: Square, Rectangle, and Triangle are 2D plane figures; Cube is a 3D solid geometry shape.

Q150. If '+' means '×', '-' means '÷', '×' means '-', '÷' means '+', evaluate: 10 + 5 - 2 ÷ 4 × 3. (Q150)

A) 26 **B)** 24 **C)** 28 **D)** 22

Correct Answer: Option (A) — 26

Step-by-Step Solution: Rewrite with symbols: 10 × 5 ÷ 2 + 4 - 3 = (50 ÷ 2) + 4 - 3 = 25 + 4 - 3 = 26.